

Aklatan+ — Proposal for Guidance & Partnership

A Hybrid Community Library System that helps public libraries modernize by integrating traditional library operations with digital library technology — designed to complement existing services and preserve the vital role of physical libraries.

Prepared for: The National Library of the Philippines (NLP) **Prepared by:** Martin Jaycy Halum — Software Developer, Rizal (10+ years, mobile & web) **Contact:** martinjaycyhalum@gmail.com · 0976-394-8909 **Date:** 2026-07-16 **Status:** Concept submitted for guidance & feedback — not a funding request

Figures marked [indicative] are placeholders for planning discussion only; they are not quotations.

1. The opportunity

Millions of students, teachers, parents, and lifelong learners depend on public libraries, yet access is limited by:

- **Distance and hours** — a reader far from a branch, or working during opening hours, is effectively cut off from the collection.
- **Physical scarcity** — a single copy serves one reader at a time; popular titles have long waits.
- **Fragmented experience** — discovery, borrowing, reading, and due-date reminders live in different places (or nowhere).
- **Uneven connectivity and devices** — solutions that assume fast internet and modern hardware exclude the very people public libraries exist to serve.

The gap is not content — the country already owns it — it is **reach and convenience**. Aklatan+ unifies physical and digital collections in one place a member can reach on the web and on their phone, **while preserving and strengthening the role of physical libraries**. Local libraries keep full ownership of their collections and their readers.

2. Our posture: complement, not replace

Aklatan+ is offered as a **practical, sustainable contribution** to public library modernization — **not** a replacement for the National Library, its partner institutions, or existing initiatives. It is built to **complement existing library systems** and to **align with the National Library's standards and long-term direction**. The concept was first presented to a local government unit as an enhancement to its e-library initiative; that experience motivated seeking the National Library's guidance to ensure alignment with national standards.

In spirit, Aklatan+ carries forward the vision of the landmark **2004 Philippine eLib** project — access to information for life-long learning, enriched local digital content, and a nationwide library network (see §7) — and aims to help realize that vision with today's technology, in cooperation with the institutions that have long advanced it.

3. What Aklatan+ offers

- **Unified management** of physical and digital library collections.
- **Mobile and web access** for library patrons, with a **home-library** affiliation so a national platform still feels local.
- **Secure digital borrowing** of eBooks and learning materials — with **in-app reading**; physical-book reservation for pickup at a member's home library.
- **A personalized dashboard** — continue-reading progress, items due soon (with clear overdue indicators), and in-app notifications.

- **Automated notifications** for due dates and material availability.
- **Borrowing history and account management.**
- **Library analytics and administrative reporting.**
- A **scalable architecture** adaptable by public libraries, schools, and local government units.

The "nationwide, still local" model

National digital content is common to all members, while a lightweight **home-library** affiliation keeps the experience rooted in each reader's local branch. This is the core design decision that lets one platform serve the whole country without erasing local libraries.

4. Where it stands today — not a paper concept

A working cross-platform build already exists: a web app and a mobile app sharing one design system, backed by managed cloud infrastructure. The member dashboard — personalized greeting, featured titles, continue-reading, due-soon alerts, in-app notifications, and home-library hours — is built and running on both platforms today. This proposal seeks endorsement to move from build to **supervised pilot**, not funding for an idea on paper.

5. Guiding principles (governance)

Every design and development decision is bound by five non-negotiable principles:

1. **Accessibility First** — WCAG 2.1 AA baseline; usable on low-cost devices and poor connections.
2. **Inclusivity for All** — plain language, multi-language-ready, no assumed technical skill.
3. **Purposeful Innovation** — technology (including AI) only when it serves a real, demonstrated need; start simple.
4. **Security & Privacy by Design** — minimal data, encryption, least-privilege, and the confidentiality of what people read treated as non-negotiable.
5. **Community Service & Sustainability** — enhance, not replace; respect intellectual property; remain affordable and maintainable for future generations.

6. Scope

In scope (initial release)

Member accounts and authentication · catalog browse/search across eBooks and physical titles · eBook borrowing and in-app reading · physical-book reservation · personalized member dashboard · home-library affiliation and branch information · in-app and push notifications.

Out of scope (initial release)

Replacing or migrating any library's existing catalog/ILS system of record · automated content digitization or rights clearance · administrative/librarian back-office tooling beyond what a pilot requires · payment or fines processing.

Assumptions & dependencies

Participating libraries can provide catalog and membership data under a data-sharing agreement · each member is associated with one home library · content made available digitally is appropriately licensed by the library.

7. Relationship to the Philippine eLib (elib.gov.ph)

Historical facts are drawn from the eLib project's own documentation and independently cross-checked against *elib.gov.ph* and *web.nlp.gov.ph* on 2026-07-14. Aklatan+ is framed as a **modern spiritual successor** — carrying forward the eLib's 2004 vision, not replacing it.

The **Philippine eLib** is a collaborative national digital library project operated by five partner institutions — **NLP, UP, DOST, DA, and CHED** — hosted at the U.P. Diliman University Library, launched via a **MOA signed February 4, 2004** with a **Php 166,770,000.00** one-year implementation budget. It is primarily a **scholarly discovery portal and union catalog**: search, subject browsing, a "book cart," and registration. Its lending model is to **reserve/request materials for physical pickup** — it has **no in-app eReader**.

Dimension	Philippine eLib	Aklatan+
Primary purpose	National union catalog + scholarly discovery	Everyday borrow / return / read experience for the public
Primary audience	Researchers, students, academics	Students, teachers, parents, general community
Lending model	Reserve/request for physical pickup	In-app eBook borrowing and physical reservation
In-app eReader	None	Yes — read and resume in-app
Platforms	Web portal	Web + native mobile (iOS/Android), shared design system
Personalization	Registration, saved selections	Home-library affiliation, reading progress, due-soon alerts
Accessibility	Adjustable text size	WCAG 2.1 AA baseline
Local-library context	National/partner-agency oriented	Per-branch home-library hours & availability

Aklatan+ re-commits to the eLib's founding objectives — life-long learning access, enriched local content, a nationwide library network — and adds what the legacy portal lacks: in-app borrowing and reading, native mobile reach, and an everyday account experience. It aims to **align and interoperate** with the eLib's collections and partner network wherever the National Library directs, respecting and complementing existing investment rather than duplicating it. *(Full comparison, including honest limitations and unverified items to confirm with the NLP: companion document 04-comparison-elib-gov-ph.md .)*

8. Technical architecture (brief)

Aklatan+ is a **cross-platform application** sharing one design system and one cloud backend across web and mobile, organized as a single monorepo.

Layer	Technology
Web	Next.js 15 (App Router), React 19, TypeScript
Mobile	Expo 54, React Native 0.81, React Navigation 7, React Native Paper

Shared	@elibrary/types, @elibrary/theme — one source of truth for domain types and design tokens
Auth	Firebase Authentication (email/password today; extensible to SSO/OAuth)
Database	Cloud Firestore (managed, horizontally scalable)
Serverless	Firebase Cloud Functions (notifications, server-side tasks)
Messaging	Expo / Cloud push notifications

There is no bespoke API server on the critical path — the apps use the Firebase client SDK directly, reducing infrastructure to operate and maintain, a deliberate choice serving the sustainability principle.

Security & privacy — reader privacy is enforced at the data layer, not merely in the client: owner-only access to borrow records, notifications, reading progress, and profile (via Firestore security rules); least-privilege writes; data minimization; encryption in transit and at rest; only appropriately licensed content is surfaced.

Accessibility — WCAG 2.1 AA baseline; keyboard-operable with visible focus (web); screen-reader labels and live-region announcements; status never conveyed by color alone; graceful degradation on low-cost devices and low bandwidth (each dashboard panel loads independently so one slow source never blanks the screen); user-facing copy centralized for future localization.

Integration — Aklatan+ complements existing library systems; it is not each library's system of record. Participating libraries keep their existing catalog/ILS. Per-library integration is via data sharing (catalog + membership) under agreement — the exact mechanism is a pilot deliverable agreed per library. No migration is required to begin.

Known gaps to close for the pilot — notification-feed generation and reading-progress persistence wired to live data sources; data-sharing integration per library; formal accessibility audit and security review sign-off; production hardening and monitoring. (*Full brief: companion document [02-technical-architecture.md](#).*)

9. Implementation roadmap

Delivery is **phased and incremental**: each phase produces something demonstrable and independently valuable, so the National Library can review and decide at each gate before committing further.

Phase	Name	Duration <i>[indicative]</i>	Primary outcome	Exit gate
0	Endorsement & agreements	2–4 weeks	Data-sharing agreement, point of contact, success metrics agreed	Signed agreement
1	Pilot readiness	4–6 weeks	Live data wiring, security review, accessibility audit, production hardening	Security & a11y sign-off
2	Supervised pilot	8–12 weeks	<i>[N]</i> participating libraries live with real members	Pilot metrics reviewed
3	Evaluation & iteration	3–4 weeks	Findings, fixes, rollout recommendation	Go/no-go decision

4	Phased nationwide rollout	ongoing	Onboard libraries in waves	Each wave stable
---	---------------------------	---------	----------------------------	------------------

Success metrics (baselined during the pilot): first-borrow success rate, time to resume an in-progress book, WCAG 2.1 AA conformance, change in due-date-related support requests, usability on low-bandwidth connections, and core-flow availability.

Indicative cost structure — one-time pilot costs (integration engineering, data services, accessibility audit, security review, production setup, enablement) and recurring operating costs (managed cloud platform, storage/bandwidth, maintenance, monitoring) are usage-based, so cost **scales with adoption** rather than requiring large fixed capacity up front. All amounts are placeholders pending real quotations. (*Full breakdown: companion document `03-roadmap-and-budget.md` .*)

Sustainability — affordable to operate (managed cloud, no server fleet), maintainable by library staff (standard, documented stack), no lock-in of library data (the library owns catalog and reader data; Aklatan+ is a custodian), with a handover plan (documentation, runbooks, training) before nationwide rollout.

10. Risks & mitigations

Risk	Mitigation
Perceived as replacing existing libraries/initiatives	Explicitly positioned to complement, not replace; local libraries keep ownership and branding
Duplicating existing national efforts	NLP guidance sought early to align with, and build on, existing initiatives
Reader privacy concerns	Privacy by design; owner-only access enforced by security rules
Content licensing / copyright	Only licensed content surfaced; no unauthorized redistribution
Low-connectivity exclusion	Accessibility & low-bandwidth resilience are release requirements
Long-term cost/sustainability	Managed cloud with usage-based cost
Adoption	Guidance-led pilot with a few libraries before wider rollout

11. A low-risk path forward

Guidance is the ask today; adoption, if the National Library ever wants it, is deliberately structured to be a small, reversible step rather than a large commitment:

- **Nothing to build from scratch** — a working cross-platform build already exists; this is a review of a working product, not a bet on a roadmap.
- **No procurement decision required to start** — a guidance-led pilot can begin under an informal collaboration or data-sharing agreement.
- **No lock-in** — participating libraries keep their existing catalog/ILS as the system of record; stopping the pilot leaves no system to unwind.

- **Reversible and observable** — each phase ends in a defined exit gate the NLP reviews before the next phase begins.
- **The National Library sets the terms** — scope, data-sharing conditions, branding, success metrics, and (if relevant) funding/IP terms are decided by the NLP.
- **Low ongoing cost if it succeeds** — usage-based cloud model means costs scale with actual adoption.

12. The ask

Respectfully, the developer seeks the **National Library's guidance, feedback, and recommendations**:

1. **Feedback and recommendations** on how Aklatan+ can best align with the NLP's standards, existing initiatives (including the Philippine eLib), and long-term direction for public library services.
2. The opportunity to **present the concept** and provide the complete proposal and supporting materials, as this document does.
3. If the direction is found worthwhile, guidance on a **small, supervised pilot** with interested libraries — including any data-sharing terms and a point of contact.
4. Should the National Library see fit, the developer remains open to discussing **funding, sponsorship, or formal adoption** of Aklatan+ for a pilot or wider rollout. This is offered, not requested — the developer is glad to proceed on guidance alone, and any such arrangement would be entirely on terms the National Library considers appropriate, covering scope, cost, intellectual property, and long-term support.

The aim is to learn from the National Library's expertise and contribute, in a modest and practical way, to the modernization of library services across the country. Thank you for your time, and for your continued work strengthening library services and literacy across the Philippines.

Respectfully, **Martin Jaycy Halum** Software Developer, Rizal · martinjaycyhalum@gmail.com · 0976-394-8909

Appendix — companion documents

This single-file proposal condenses the full submission package. Fuller detail on any section is available on request:

#	Document	Purpose
02	Technical Architecture Brief	Full stack, data model, security, accessibility, scale detail
03	Implementation Roadmap & Budget	Full phased plan, metrics, cost breakdown, resourcing
04	Relationship to the Philippine eLib	Full historical comparison and honest limitations
—	Constitution (governing principles)	<code>.specify/memory/constitution.md</code>
—	Member dashboard specification	<code>specs/001-user-landing-page/spec.md</code>